

DEPARTMENT OF MINES
SOUTH AUSTRALIA.

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GENERAL GEOLOGICAL REPORT ON THE MT. GUNSON - PERNATTY
LAGOON DISTRICT AND MINERAL DEPOSITS.

INTRODUCTION

An area of approximately 900 square miles was examined in the course of ten days with a view to establishing the fundamental relationships between the mineral deposits and the rocks and rock structures in the area in question. The geological map accompanying this report shows the actual limits of the reconnaissance survey which was carried out, and the inset map, thereon, shows the localities of the main mineral deposits of the district.

SUMMARY OF MINERAL OCCURRENCES

The mineral deposits were found to be localized within or adjacent to a distinctive bed of dolomitic limestone. The singular expression "bed" is used because only one bed or layer of limestone was seen during the reconnaissance. The association of the mineral deposits with this bed of dolomitic limestone may be considered to hold throughout the Mt. Gunson district, but it may not hold a few miles outside the district where different conditions may prevail. The distribution of this favourable bed of dolomitic limestone is shown on the geological map and its limits fairly well define the area beyond which the chances for ore are extremely unlikely.

The actual structural controls producing the localisation of the ore deposits were not discovered in such a cursory examination. They are undoubtedly obscure and indefinite in character on account of the relatively undisturbed nature of the sedimentary rocks of the district. Surmises as to their nature are made later in reference to the origin of the mineral deposits. These deposits are believed to be chiefly of epigenetic origin and to have been formed by meteoric waters.

MICROFILMED

The mineral deposits comprise those of copper, manganese fluorite and barytes, the latter two being of little economic interest on account of their sparse occurrence. Copper is being produced at the Mt. Gunson copper mine. This mine was examined in more detail and forms the subject of a separate report. Manganese deposits are being worked on the western side of Pernatty Lagoon about 5 miles north of Woocalla. Reports on these deposits have been published in Mining Review Nos. 25 and 31.

TOPOGRAPHY

The area is composed of isolated tablelands, interspersed with extensive depressions occupied either by salt lakes or by sand dunes and alluvial swamps. The term "tableland" is used for broad elevated area bestrewn with fragments of basement rock and "Durricrust" material. On the tablelands the vegetation is very scanty with a marked absence of trees. In contrast the low-lying areas have a well developed covering of mallee trees except around the margins of the salt lakes.

A few high flat-topped residual hills stand out above the general level of the tablelands and form conspicuous features of the landscape. Mt. Gunson, South Oakden, North Oakden and West Oakden Hills are typical residual hills. Channel Hill at Wirrappa Siding, on the East-West Railway line is also a residual hill. It is adjacent to a more closely bunched series of mesa-like remnants of a former land surface which was situated about 500-1000 ft. above the present surface. The process of formation of the present flat land surface as a result of widespread degradation is well advanced.

In this district faulting in Tertiary times, is difficult to detect in terms of differential levels and in movements of large displacement may be entirely absent. Broad warping movements of the rocks have very probably taken place, however, and may be responsible for the slight variations in level of the different tablelands and of the residual hills.

GENERAL GEOLOGY

The rock succession of the area may be summarized as follows:-

1. Alluvium and wind-blown sand, either in the form of dunes or as a thin veneer, occurring for the most part in the low-lying salt basins and swamps.
2. The "Durricrust" material composed of fragments and disjointed masses of siliceous laterite, a secondary quartzite formed in Tertiary times from soils and alluvium under certain climatic conditions. The "Duricrust" material is associated with angular highly silicified fragments of the underlying rocks and occurs in the form of a mantle covering the tablelands.
3. White pipe clay material occurring at Woocalla and possibly on the south margin of Pernatty Lagoon as small inliners of rock of tentative Cretaceous Age.
4. The sedimentary rocks forming the bedrock of the region in the following order of succession from youngest to oldest.

III. Upper Sandstones.

- B. White sandstone (100ft. plus)
- A. Red sandstones and grits (600ft.)

II. Dolomitic Limestone (50-188ft.)

I. Lower sandstones (50ft. plus)

A thickness of nearly 800ft. of basement rocks is inferred from the exposures examined in this area. These rocks show all the features indicative of shallow water and terrestrial conditions of deposition.

It is difficult to assign an age to these sediments as no definite fossil evidence was found nor has it been recorded in previous reports. The South Australian geological maps up to and including the 1928 edition, assigns to them a tentative Ordovician age, but the Late Professor Sir Edgeworth David depicts them as Pre-Cambrian (Newer Proterozoic) on his geological map of the Commonwealth of Australia. The age of these rocks will be discussed in more detail in a subsequent section

of this report.

Descriptions of the various horizons of the above rock column are as follows:-

III Upper Sandstones.

The upper Sandstones are so called because they overlies the Dolomitic Limestone which forms a distinctive stratigraphic horizon in the rock succession. The Upper sandstones are subdivided into two groups:-

B. The White Sandstones

A. The Red Sandstones and Grits.

B. The white sandstones occur as hard resistant cappings on the residual hills, namely Channel Hill, South Oakden Hill, North Oakden Hill and Mt. Gunson, etc. They also occur on the higher parts of the tableland between the Birthday Siding and the Mt. Gunson Mine. Three lithological sub-groups are definable in the following order from youngest to oldest.

- iii. 40 ft. (plus) of sandstone of variable grains size, not particularly well bedded, but containing hard dense quartzite layers usually showing curious casts which strongly suggest an organic origin. This is the youngest rock horizon of basement rocks exposed in the area.
- ii 30-40ft. of white quartzitic sandstone, massive bed which stands out in bold relief on or near the top of the residual hills. Its weathered surface is somewhat ferruginous and presents a curious pitted appearance, the small cavities being on the average half an inch in diameter. On close examination it was found that the cause of this pitting was due to the presence in the rock of numerous uncemented patches of fine sand which readily disintegrated on being exposed to the weather. The bed is best seen on Channel Hill near the East-West Railway line on the south end of Lake Windabout. A conglomerate bed, 2ft. thick, containing on the average one inch diameter round sandstone and quartzite pebbles was seen at the top of this white quartzite horizon on the Mt. Gunson Mine - Birthday Siding road.

- i. 30-40ft. of white crossbedded sandstone. This rock horizon is distinguished by its well-defined cross bedded character. In places it shows cross bedding structures very similar to those of typical aeolian or dune cross bedding. The cross bedding suggests aeolian conditions of deposition for these lower white sandstone beds, but from the above facts the upper part of the white sandstone group was deposited under shallow water conditions.

A. The Red Sandstones and Grits cover the major portion of the area. They have for the most part, in weathered outcrop, a distinctive red colour. Grain size is very variable. Bedding is generally well marked with current or cross bedding universally present. Shale layers are interbedded in places with the red sandstones, but are not universally present and generally are not conspicuous anywhere in the district. The upper portion of the red sandstones 'and grits' horizon is well exposed on the flanks of South Oakden Hill, where a thickness of 300ft. can be studied. At this locality the contact between the Red Sandstones and the White Sandstones outcrops. This contact can also be seen on Channel Hill and on Mt. Gunson. It is defined chiefly by a colour change. The correlation of this contact on Mt. Gunson, Channel Hill and South Oakden Hill, at approximately the same level, is indicative of the existence of stable conditions in the region over a long period of geological time.

The lower portion of the Red Sandstones and Grits occurs in the vicinity of the Mt. Gunson Mine, where they are copper bearing and are, therefore, important from an economic aspect. These rocks are generally highly siliceous and where their copper content is sufficiently high, they form a copper ore which is amenable to treatment at the Port Pirie Smelters, where a siliceous flux is used. The ore occurrences in these rocks are discussed further on in this report

II Dolomitic Limestone.

This rock horizon has a variable thickness. It may or may not have a continuous character throughout the region. Its occurrence, however, is sufficiently extensive to assume, with fair precision, that it forms a persistent stratigraphic horizon with which the general geology of the district can best be illustrated. The configuration of the dolomitic limestone bed is of great importance, because all the manganese and copper deposits are distributed either on it or in the sandstone beds immediately above or below it. Phosphates are recorded as having been found in joint planes in the limestone, but the occurrences are of no economic value.

The rock has a grey colour and has a blocky, jointed appearance. It represents recrystallized carbonated rocks. In the northern part of the area at Mt. Gunson Mine it has a thickness of about 50ft., whilst in the southern part of the area, in a bore about 8 miles north of Narkanna Well, it is over 188ft. thick. It is well exposed in the Commonwealth Railway Ballast Quarry at Woocalla siding, where it appears to overlie conformably the Lower Sandstones.

It is quite possible that this contact may represent a geological time break but no evidence was obtained to support this contention. In Mining Review 25, p. 55, the Government and Assistant Government Geologists state that the limestone contact in the Woocalla Quarry is an unconformable one. Structural evidence cannot support such a judgement as strikes and dips in sandstone and limestone are conformable.

The contact of the dolomitic limestone and the upper sandstones was not seen. It is reported in certain of the Mt. Gunson Mines workings which were inaccessible at the time of the inspection. Dips and strikes in the dolomitic limestone and overlying sandstones are similar and conformable relationship between the two rock horizons can be at least safely assumed for structural and ore occurrence consideration.

From a stratigraphic point of view a disconformity may exist between the dolomite and the Upper and or the Lower sandstones but its possible occurrence cannot be supported by facts.

I The Lower Sandstones.

The lower sandstones are red in colour and are lithologically similar to the Red Sandstones of the Upper Sandstone series. Those rocks may be seen in the Commonwealth Railway Ballast Quarry at Woomalla, where a thickness of them of about 50ft. is exposed and where the contact with the overlying dolomitic limestone was observed. At the Sweet Nell Mine, on Pernatty Lagoon, these lower sandstones are cupriferous immediately below the dolomitic limestone. Shaley layers occur in the lower sandstones on the western margin of Pernatty Lagoon near Mt. Gunson, but shales are nowhere the predominant rock type. By and large, only a small thickness of the Lower Sandstones (say 50ft.) outcrops in this district.

5. The rocks at depth below the badement sediments.

These rocks are of interest because they are the most probable sources of the mineral deposits of the Mt. Gunson Pernatty Lagoon District. At localities near the district in question in deep bores, sunk for water, igneous rocks have been intersected. To the south and south west porphyritic and granitic rocks outcrop in the region of Iron Knob and the Gawler Ranges. Slates are known to occur in the Lake Torrens area.

The older rocks at depth, therefore, can reasonably be cupriferous. The possible occurrence of granite, porphyry and slates suggests an excellent environment suitable for the development of copper deposits.

REGIONAL GEOLOGICAL STRUCTURE

The Upper Sandstones, the Dolomitic Limestone and the Lower Sandstones form an apparently conformable series of sediments which have a subhorizontal arrangement.

Broadly speaking the structural setting throughout the region is one of a gentle basin and cone-like structure, locally fractured and faulted.

The lenticular character of the beds and the prevalence of current bedding frequently gives an impression locally, of the occurrence of folds, but careful inspection showed the error of such conclusions. It was only by extensive mapping that any general warping of the beds with minor local fracturing was revealed. No true dips exceeding 25° were observed.

The regional structure was satisfactorily determined in spite of the heavy accumulations of overburden by tracing, throughout the district, the general behaviour of the dolomitic limestone. The dolomitic limestone serves as an excellent marker horizon or key bed. In consequence, throughout the district, its interpolation in plan and section as a theoretically continuous bed provides a simple and clear means of illustrating the regional structure of the district.

The persistent character of the dolomitic limestone brings the assumption of the continuity of this bed very close to actuality. The association of the mineral deposits with the dolomitic limestone make the interpolation of this key bed all the more important, because at any locality its position gives the favourable rock horizon for likely ore occurrences. The dolomitic limestone key bed has been accordingly accentuated on the interpolated plan and sections accompanying this report.

No definite evidence of the occurrence of faulting in the district was obtained in the course of the reconnaissance. However, it would be wrong to surmise that faulting is entirely absent from the district. The region is adjacent to the Spencer Gulf-Lake Torrens meridional rift region in which major block faulting has occurred. It is quite probable therefore, that minor supplementary faulting has occurred in this district. This contingency is not indicated on the interpolated geological maps, owing to the dearth of factual evidence. The failure to locate definite faults in the district makes the establishment of a genetic relationship between concentration of minerals and

geological structure, an extremely difficult matter.

GEOLOGICAL AGE OF THE BASEMENT ROCKS

The basement rocks of the Mt. Gunson - Pernatty Lagoon region have failed to yield any fossils with which an age can be assigned with fair precision to the rocks.

On lithological grounds, they were initially given a tentative Ordovician age and were represented on the geological map of South Australia as such.

In Bulletin 18, G.S.S.A., Mr. R. W. Segnit shows on Fig. 63, p. 173, the Mt. Gunson - Pernatty Lagoon sediments belonging to the Upper Pre-Cambrian portion of the Lower Cambrian - Pre-Cambrian system of sediments which has a wide distribution in South Australia. This age has been given on a correlative basis, as Mr. Segnit has not mentioned the Mt. Gunson area, in particular in that Bulletin.

The rocks of the Mt. Gunson - Pernatty Lagoon area are contiguous with the rocks extending both north and south of the area and form part of one extensive relatively undisturbed sedimentary series. This series is assumed by Mr. Segnit to form part of the Pre-Cambrian - Cambrian system so well developed on the eastern side of Spencer Gulf and Lake Torrens. To accept this assumption in order to find the position of the Mt. Gunson - Pernatty Lagoon basement rocks in this system, a correlation of the rock column of that district with a section of the complete stratigraphic succession of the system must be made. Such a correlation, however, is difficult to establish, as will be seen in the following considerations, and, therefore, the age of the Mt. Gunson rocks remain very uncertain. The whole problem of the geological age of the rocks is chiefly concerned with the determination of the position of the dolomitic limestone in the system.

For the purposes of correlation, the complete stratigraphical succession of Cambrian and Pre-Cambrian rocks, outlined by Mr. Segnit in Bulletin No. 18 G.S.S.A. p. 188-9, is reproduced below, together with the Mt. Gunson - Pernatty Lagoon rock column.

It will be required for reference purposes in the following discussion.

I. GENERALIZED STRATIGRAPHICAL SUCCESSION OF THE GEOLOGICAL FORMATIONS OF THE LOWER CAMBRIAN - PRE-CAMBRIAN SEDIMENTARY SERIES.

(Bull. 18, G.S.S.A. p. 188-189).

Lower Cambrian -

Calcareous Sandstone	E	
Archaeocyathinae Limestone	}	E
Fossiliferous Limestone		
Grey to purple Limestone	E	- - - - - (1)
Shales	E	

Disconformity, (or unconformity ?)

Upper Pre-Cambrian -

Flinders Range Sandstone Quartzite Series D7

Purple Series -

Upper Group -

Purple slates D6h

Middle group -

Sandstone	}	D6b - g
Quartzite - Sandstone Conglomerate		
Siliceous Slates and Quartzites		
Dolomitic Limestone - - - - - (2)		
Grey Shales		
Felspathic Sandstone		
Quartzite		

Lower Group -

Purple Shales D6a

Dolomitic Limestone D5 - - - - - (3)
Dolomitic Quartzite D4

Tapley's Hill Series -

Upper group-

Grey Slated D3g

Dolomitic Limestone D3f - - - - - (4)

Middle Group -		
Dark Grey Slates	- - - - - (5)	} D3b - e
<u>Grey Limestone</u>		
Blue-Grey Calcareous Slates and Shales		
Grey Quartzite		
Lower Group -		
Calcareous (Ribbon) Shales		D3a
<u>Post - Glacial Dolomitic Limestone</u>		D2 - - - - - (6)
Disconformity.		
Sturtian Tillite -		
Upper Boulder Beds		D1c
Intercalated Series (Quartzite, Dolomitic Limestone, Shales and Tillite)		D1b
Lower Boulder Beds		D1a
Unconformity.		
Middle Pre-Cambrian -		
Upper Series -		
Felspathic Quartzite		C5
Ferruginous Felspathic Sandstones		C4
<u>Dolomitic Limestones</u>		C3 - - - - - (7)
<u>Quartzites</u>		C2
Shales		C1
Lower Series -		
<u>Dolomitic Limestone</u>		CL - - - - - (8)
Shales and Slates		CL
Quartzite		CL
Argillaceous Sandstone		CL

It will be seen in the above generalized succession, Table I, that limestone occurs in eight different parts of the succession. These limestone occurrences are underlined and numbered. They will be correlated in turn with the Mt. Gunson occurrence.

II MOUNT GUNSON - PERNATTY LAGOON SEDIMENTARY SUCCESSIONS.

Upper Sandstones.

White Sandstone (100ft. plus)
Red sandstones and grits (600ft)

Dolomitic Limestone (50-188 ft. plus)

Lower Sandstones (50ft. plus)

1. If the Mt. Gunson limestone is given a Lower Cambrian age corresponding to (1) in the above table, the underlying sandstones can reasonably be inferred as Upper Pre-Cambrian Flinders Range Sandstone - Quartzite Series, but the sandstones overlying the dolomitic limestone at Mt. Gunson find no correlative in this portion of the stratigraphic succession.

2. If the dolomitic limestone of Mt. Gunson is made equivalent in age to the dolomitic limestones of the Middle group of the Upper Pre-Cambrian Purple Series (2) above, the sandstones both above and below the limestone at Mt. Gunson find no close correlatives in the succession owing to the preponderance of shales, in general, in the Purple Series.

3. If the Mt. Gunson limestone is made equivalent in age to the Upper Pre-Cambrian dolomitic limestone below the Purple Series (3 above), the overlying sandstones of Mt. Gunson cannot be correlated with the Purple Shales in a comparable position in the general rock succession.

4 & 5. If the dolomitic limestone of Mt. Gunson is given a similar age to that of the Tapley's Hill series (4 and 5 above), the correlation is questionable, because the latter series have a predominant development of shales both above and below the limestone horizons.

6. As no sign of Tillite occurs at Mt. Gunson beneath the limestone, or of calcareous shales above it, the Mt. Gunson limestone cannot be correlated with the Post Glacial Dolomitic Limestone (6 above) of the generalized succession.

7. With the Middle Pre-Cambrian limestone (7 above), the best correlation is made between the Mt. Gunson rocks and those outlined in the generalized succession. The Upper and Lower Sandstones and Grits at Mt. Gunson are comparable to the sandstones and quartzites respectively in the generalized succession.

8. If the limestone of Mt. Gunson is made lower Middle Pre-Cambrian in age similar to (8) of the generalized succession, the correlation of the Mt. Gunson sandstones with comparable beds in the generalized section is difficult owing to the presence of shales both above and below this horizon in the generalized succession.

The net result of the above line of enquiry suggests the following possible ages for the Mt. Gunson rocks in the Lower Cambrian-Pre -Cambrian System:-

A. The Mt. Gunson limestone is Lower Cambrian. The Upper sandstones at Mt. Gunson either conformably overlies the limestone and are of Cambrian age, or they disconformably overlies the limestone and may be of Post-Cambrian age. They may quite possibly be Tertiary! Lithologically, however, they are markedly similar to the Lower Sandstones which would belong, on the above assumption to the Flinders Range Sandstone - Quartzite Series of Upper Pre-Cambrian age and it is, therefore, likely that the disconformity, if it exists, between the limestone and upper sandstones at Mt. Gunson only represents a small geological time break.

B. The Mt. Gunson limestone belongs to the Middle group of the Purple Series of Upper Pre-Cambrian Age (D6 b - g). This age is considered possible because the overlying and underlying sandstones with their occasional thin shaley layers have a red tinge, - secondly, because there occurs quartzites and sandstones in this section of the generalized rock succession. It would be erroneous to neglect this possibility on account of the absence of shales in the Mt. Gunson rock column. Lenticularity of beds must be allowed for, as well as the existence of changing conditions of deposition from place to place.

c. The Mt. Gunson limestone is of Middle Pre-Cambrian age. This age as indicated previously, fits all the facts presented by the rock column at Mt. Gunson and by the generalized succession, and, therefore, appears to be the most likely as a result of this line of enquiry.

However, it is wrong thinking to base age determination solely on hard and fast correlations depending on lithological qualities of rocks. None of the above suggested ages can be accepted without further considering the regional geology of the Mt. Gunson - Pernatty Lagoon District. To the north of Mt. Gunson at Andamooka, Roxby Downs and Parakylia Stations, sandstones of similar character to those at Mt. Gunson have been assigned an Upper Pre-Cambrian age and have been correlated with the Flinders Range Sandstone - Quartzite Series by Mr. Segnit. To the south at Lincoln Gap, a similar age has been assigned to similar sedimentary rocks. The continuity of these sediments

from Lincoln Gap to the country north of Mt. Gunson is an established fact. The Mt. Gunson - Pernatty Lagoon sandstones in consequence most probably form part of the Flinders Range Sandstone - Quartzite Series. This line of reasoning leads to a fourth possible age for the rocks of the Mt. Gunson Region, namely, Upper Pre-Cambrian (D7) and it is based on acceptance of Mr. Segnit's conclusions concerning similar rocks in the same geological province.

Further, the acceptance of this Upper Pre-Cambrian age introduces the occurrence of limestone into the Flinders Range Sandstone - Quartzite Series, 7 (d), an occurrence which has not been reported to date in this series.

The above analysis suggests four possible ages for the Mt. Gunson rocks and for the limestone in particular if these rocks are members of the Cambrian Pre-Cambrian succession.

1. Lower Cambrian (E)
2. Upper Pre-Cambrian (D7)
3. Upper Pre-Cambrian (D6 b - g)
4. Middle pre-Cambrian (C3)

The present uncertainty of the geological age of the Mt. Gunson rocks is due either to the failure by previous investigators to take into account the dolomitic limestone occurrence at Mt. Gunson, or the limestone occurrence has not been critically considered in regard to its possible age prior to this report.

In Bulletin 14, G.S.S.A., Dr. R.L. Jack has drawn a geological section passing through the southern end of the Mt. Gunson district. On the section Dr. Jack shows Gibson's Camp Bore, near Narkanna well and he has inferred the hard blue rock in the bottom of this bore to be slate. In recent drilling for water on Oakden Hill Station, at a locality about eight miles north of Narkanna Well, limestone was intersected in a bore at a vertical depth of 300ft. The bore reached a total depth of 488ft. without passing out of the limestone rock.

The result of this drilling suggests that the hard blue rock in Gibson's Camp Bore is limestone and not slate as Dr. Jack has inferred it to be. If this modified interpretation is correct, the slates, found in the neighbouring country to the east of the

Mt. Gunson district are at a much greater depth in the Mt. Gunson district than was formerly believed. This line of reasoning suggests, therefore, that slates underlie the Lower Sandstones in the Mt. Gunson District and by correlation with the generalized succession above, these slates may be Middle Cambrian (C1) in age thereby marking the overlying Lower Sandstones at Mt. Gunson Middle Cambrian (C2), the Dolomitic Limestone (C3) and the Upper Mt. Gunson Sandstones, Middle Cambrian (C4).

The results of investigation into the age of the Mt. Gunson rocks are therefore very indefinite.

GENERAL CONSIDERATION OF THE MINERAL DEPOSITS

A. The Copper Deposits. The copper deposits which have been worked in the past, in the district, include those at Mount Gunson Mine, Pernatty Lagoon, Gunyot, Ramsay, Gun and the Sweet Nell. The Mount Gunson deposit at the present moment is the only one being worked and it will be considered later in detail.

The copper in general occurs in sandstone beds which are in contact or in close stratigraphical proximity to the dolomitic limestone which also contains copper minerals. The copper minerals are found in the sandstones in very restricted areas, but in the dolomite, traces of copper minerals can be seen having a more wide-spread distribution. The workable deposits up to the present have been restricted entirely to the sandstones. The Mt. Gunson deposit occurs in the Upper Sandstones, whilst the Sweet Nell deposit occurs in the Lower Sandstones. The copper minerals are distributed in the sandstone beds in a very irregular fashion. This irregularity is accentuated by the heterogeneous character of the beds in which lenticularity and cross bedding are most marked. The thinner sandstones, comparatively softer and more friable tend to be more genial towards receiving the copper minerals. These friable bands, along strike, frequently change to massive quartzite sandstones with often an appreciable diminution of mineral content.

The following description of the nature of the mineralization at Mt. Gunson was given by H.J.C. Conolly in a report on the mine to the Zinc Corporation:-

"Oxidized ores of copper comprising chloride sulphate, carbonate and siliceous minerals, either as individuals or admixtures of the amorphous type, together with chalcocite or copper glance, occur within the lower beds of the Upper Sandstone. In many places the green copper minerals are developed throughout the sandstone beds forming a cement to the sand grains whilst in other exposures, they are confined to joints, cracks and bedding plane partings. In some sections both conditions of occurrence are present. Chalcocite is exhibited as veinlets filling cracks and as blebs within the sandstones with or without the oxidized minerals. The more friable and gritty sandstones appear to contain the oxidized minerals in a more evenly distributed form than the denser fine-grained quartzitic beds."

"That the copper contents are distributed irregularly is apparent from an examination in the Mt. Gunson Open Cut and Tunnel. A full section of five to six feet in thickness may be seen to merge gradually into barren sandstone or, on the other hand the lower beds may become lean and the higher persist as a copper-bearing horizon. No uniform behaviour can be established, but it is quite obvious that barren and lean masses must be expected as the beds are followed in any direction."

These descriptions apply equally well in the case of the Sweet Nell deposit where the mineralization is developed in sub-horizontal sandstones below the Dolomitic limestone over a width of about 3ft.

At the Pernatty Lagoon workings about 2 miles north of the Mt. Gunson Mine, the copper minerals are principally sulphides, chalcocite, covellite, bornite, with subordinate atacamite, malachite, etc. This occurrence could not be examined because all the workings have either fallen in or were filled with salt

water. The occurrence is of interest because it represents a transitional stage in the process of oxidation of a bedded sulphide ore body. Mt. Gunson is a completely oxidized deposit, whereas the Gun deposit, half a mile east of Mt. Gunson, according to Mr. W.E. Wainwright is a partially oxidized deposit, possibly in a later stage of oxidation to that of Pernatty Lagoon sulphide deposit. The report of W.E. Wainwright on Pernatty Lagoon, Gun, Gunyot and Ramsay deposits appears in Mining Review No. 20 pp 41-51. The Gunyot deposit is similar in character to the Mt. Gunson deposit.

The following grouping, therefore, summarized the relative state of oxidation of the copper deposits of the district.

Mt. Gunson, Gunyot, Sweet Nell - completely oxidized.

Gun - not completely oxidized.

Pernatty Lagoon - slightly oxidized.

B. The Manganese Deposits.

Manganese is the most commonly occurring mineral in the Mt. Gunson district. It can be detected in the hand specimen, in the dolomitic limestone, in practically every locality where this rock outcrops. Only in a few places, however, is the concentration of manganese mineral sufficiently great to form workable ore deposits. The deposits which are being worked at present occur on Pernatty Lagoon about half way between Mt. Gunson and the Sweet Nell copper deposits, and about 5 miles north of Woocalla. The manganese ores exploited to date are solely oxidized ores composed mainly of manganese oxides, chiefly pyrolusite. Barite and fluorite have been reported as occurring in close proximity to the manganese deposits, whilst barite and calcite are known to occur as minor accessory minerals in the manganese ores. The crushed ore produced contains approximately 35-45% manganese.

The manganese ore occurs in decomposed broken zones in the dolomitic limestone. In some places large bodies of ore are formed with ill-defined boundaries. In other places large patches of little altered or unaltered dolomitic limestone are entirely surrounded by ore. In the short time spent as the deposit there was no opportunity to seek a structural explanation of the deposit. The ore and dolomitic limestone is brecciated in certain localities in the deposit and the brecciation suggests

there may possibly occur a fault or fault pattern or zone which would provide the passages for the ore-forming solutions and facilitate the replacement of the limestone. To test this hypothesis it would be necessary to study the sandstone dolomitic limestone contacts and look for any possible offsets of them attributable to faulting.

Owing to the extremely erratic distribution of the manganese minerals in the ore zone, it would be very difficult to lay out a testing programme for ore reserve estimates without a careful study of the structure and the peculiarities of the ore bodies. Little ore, at present is showing beyond working faces, but there are favourable indications of further ore and development and exploratory work should be further advanced as the possibilities of the deposits are most promising. Earlier and more comprehensive reports on the deposits appear in Mining Reviews Nos. 25 and 31.

Elsewhere in the district small prospecting workings have opened up small patches of manganese ore, but by no means have the possibilities of the district been fully explored.

Origin of the Mineral Deposits.

Bedded mineral deposits, particularly those of copper have been the subject of considerable study throughout the world in regard to their mode of origin. The study of the Mt. Gunson - Pernatty Lagoon mineral deposits offers further enlightenment on this problem in that it involves a consideration of bedded copper occurrences in various stages of oxidation and at the same time includes the problem of an interesting mineral association, namely Copper and Manganese. The deposits of the two minerals have many features in common and in consequence appear to have comparable modes of origin.

The following chemical analysis of the dolomitic limestone shows the association of copper and manganese.

The sample was taken some distance away from worked mineral deposits.

Mr. T.W. Dalwood, assayer to the Mines Department, who made the analysis, described the copper as occurring with the manganese in fine veins throughout the mass.

Silica (SiO_2)	1.42%
Alumina (Al_2O_3)	0.74
Ferric oxide (Fe_2O_3)	1.34
Lime (CaO)	28.99
Magnesia (MgO)	21.12
Water (H_2O)	0.50
Carbon dioxide (CO_2)	44.28
Cupric oxide (CuO)	0.21 (0.17 Copper)
Manganese oxide (MnO)	<u>1.92 (1.49) Manganese)</u>
	<u>100.52%</u>

The following treatment presents a purely hypothetical picture of the possible genesis of the mineral deposits of the Mt. Gunson - Pernatty Lagoon district, because it is based entirely on facts observed in the field.

The main features of the mineral occurrences are as follows: -

1. The deposits are far removed from igneous rocks and, therefore, the source of the minerals is quite obscure.
2. The ore bodies occur chiefly as metasomatic replacements and partly as fillings in the form of concretions along bedding and fracture planes in the sedimentary rocks.
3. The ore bodies are restricted to one particular stratigraphic horizon.
4. The mineralization is for the most part, particularly in the case of copper, restricted in area.
5. The mineralization does not follow in all cases, the bedding planes or one definite sedimentary rock type.

Three clear-cut hypotheses have been advanced to explain the origin of this type of mineral occurrence. Firstly, they may be of syngenetic origin, in which the minerals in the deposits are conceived as being concentrated at the time of deposition of the beds. The most serious objections to this hypothesis are the following:-

- (i) The ores are not restricted to one single rock type, such as limestone.
- (ii) The ores are not confined to one definite bed of rock.
- (iii) Finally, there is an absence of possible precipitants, e.g., traces of plant remains, etc.

The second hypothesis suggest a hypogenetic origin in which the minerals are conceived as being deposited by ascending hydrothermal solutions. This hypotheses is quite acceptable but it does not adequately consider all the facts and there is one serious objection to it, namely, there does not exist any strong evidence at present of a definite genetic relationship between geological structure and ore occurrence, a relationship which is an essential basis for such an hypothesis.

The third and final clear-cut hypothesis is the epigenetic origin enunciated by Lindgren in which "the ores have been concentrated by meteoric waters, which leached the small quantities of metals disseminated in the strata." The objection to this hypothesis is also the fact that to date an ore-structure relationship has not been established. This failure must be at present attributed to an insufficient amount of geological study having been done in the region. It would, therefore, be erroneous to incline towards an acceptance of the non-existence of a definite ore-structure relationship.

In spite of the lack of definite structural facts, the third or epigenetic hypothesis is the most feasible at any rate for copper deposits. When the occurrence of manganese is considered in conjunction with that of copper, one hesitates to pronounce this hypothesis as holding throughout the district because manganese is common mineral of ores of magmatic origin. Also the reported occurrence of dulongite and barite raise further doubts. They suggest that a hydrothermal origin, at least in part, cannot be entirely eliminated from the picture of ore genesis. At any rate the presence of these minerals indicates that the ore solutions were hot rather than cold.

Generally, however, the epigenetic mode of origin is considered dominant. The copper deposits are typically "Red Bed" occurrences. The copper minerals are conceived as having been derived from copper bearing lodes of disseminated copper masses in the older rocks at depth. It has already been suggested that these rocks can reasonably provide the necessary conditions for mineral deposits in them. The copper minerals were leached from these deposits by heated saline meteoric waters. These waters penetrated the older rocks through fissures. The copper minerals were transported in solution by meteoric waters to certain loci of deposition in the overlying porous sandstones. Deposition was probably brought about in part by the action of reducing agencies, the natures of which are obscure. It is also very probable that the dolomitic limestone acted indirectly in the precipitation of copper. For example, the limestone may have acted in the capacity of a relatively impermeable bed and the establishment of favourable physical rather than chemical conditions of precipitation.

H.J.C. Conolly and J.K. Gustafsen, who examined the Mt. Gunson mines for the Zinc Corporation, favour this epigenetic mode of origin, at least, for the Mt. Gunson deposit. Conolly has reported evidence of faulting at Mt. Gunson, and although it was not seen by the writer, the following extract from Conolly's report adds further strength to the validity of the above hypothesis. "The distribution of the copper contents in the sandstones appear to be related to fissures formed by minor faulting. There is evidence of a fissure in the open cut near Brennan's Shaft, and it is suggested that there is some probability that the location of the various deposits have been determined by infiltrations of solutions from channels formed by such faults, in contradistinction to a more or less uniform distribution, throughout the sandstone beds. That is, the copper content have been introduced subsequent to the accumulation of the sediments and may be more enriched in proximity to the fissures".

In this respect Lindgren believes epigenetic ores to have been forming continuously since the establishment of active water circulation in the beds.

The initial copper mineral formed would very probably have been chalcocite, which would then be subject to alteration depending on the level of ground water. The Pernatty Lagoon deposit occurring at and below the present water level, contains mainly chalcocite. On the other hand, the Mt. Gunson deposit contains only traces of chalcocite. The major portion of the original chalcocite in the deposit has undergone decomposition and oxidation owing to a lowering of the water level subsequent to the deposition of the chalcocite.

E.S. Bastin (Econ.Geol. vol. 28, p. 422) suggests that bedded copper deposits may in places be subjected to a cycle or cycles of changes involving deposition, erosion, oxidation and solution, transportation, and redeposition before they attain their present form. Such a conception is not tenable here, because the copper minerals are related solely to one stratigraphic horizon and the configuration of the present land surface of the ground water table is not controlling factor of the loci of deposition. The position of the ground-water table only controls secondary enrichment as it does in the majority of mineral deposits.

The genesis of the manganese deposits is very probably similar in most features to that of the copper deposits on account of their comparable characteristics of the respective deposits. The manganese is, therefore, conceived as originating from mineral deposits at depth, the manganese being transported in solution by meteoric waters through channels produced by faulting and fracturing. On coming in contact with the dolomitic limestone, these solutions encountered favourable conditions for the precipitation of the manganese minerals.

The extent of secondary enrichment in the manganese deposits is not known. Enrichment caused by the concentration due to the natural erosion and solution of the dolomitic limestone is in residual clays containing patches and veins of manganese. The clays are formed from the weathering of the dolomitic limestones containing manganese.

ASSISTANT GOVERNMENT GEOLOGIST

REPORT ON THE MOUNT GUNSON COPPER DEPOSITS.

(Supplementary to the General Geological Report on the Mount Gunson - Pernatty Lagoon District).

LOCATION

The Mount Gunson copper mine is situated near the western margin of Pernatty Lagoon, 12 miles north east of Birthday Siding on the Transcontinental railway line, Birthday Siding is about 80 miles from Port Augusta at the head of Spencer Gulf. From the Siding to the mine there runs a graded road which intersects the main Port Augusta - Tarcoola road 3 miles distant from the Siding.

CHARACTER OF THE COPPER DEPOSITS

The general character of the Mount Gunson deposits has been described in the geological report on the Mount Gunson Pernatty-Lagoon District.

The known copper ores occur as flat, irregular tabular bodies in sub-horizontal sandstone beds immediately overlying a dolomitic limestone. The vertical range of the ore horizons in the sandstones would not exceed 30ft. The widths of the individual ores seams vary from 30in. to 132in. (plus). They have a strike east of north and dip northward at a flat angles. The strike and dip of the copper occurrences conform very closely with those of the sandstone.

Two ore horizons or seams appear to have been developed and a third is suggested but not proved. They are as follows (see map):-

- (1) The main ore horizon in the Mount Gunson open cuts.

This horizon is known to extend south westward through the Tableland shaft to Brennan's workings. It follows very closely a contact between a red sandstone and a white sandstone, the latter one overlying the former. The copper horizon in the Smelter shaft is probably on the extension of the main ore horizon going north from the main open cut (see section map). This

conclusion is based on the assumption that the main copper horizon has a flat northerly dip (say 5-10 degrees) between the main open cut and the Smelter shaft. Dr. J.K. Gustafsen, on the contrary, believes the Smelter shaft horizon to be a separate horizon, which is stratigraphically below the main open cut horizon. Both these contingencies should be allowed for until further testing or development work supplies additional data.

(2) The ore horizon in Brennan's shaft.

This horizon occurs in a sandstone bed immediately overlying the dolomitic limestone. Until further information is available, the correlation of it with (1) above, or with the Smelter shaft horizon is inly guess work. For the present this horizon is considered to be a separate horizon stratigraphically below the main open cut horizon (1).

(3) An ore horizon may occur immediately below the dolomitic limestone in a position comparable to that of the Sweet Nell ore-body. Up to the present no exploratory work has been done to test such a possible ore position.

EARLY HISTORY OF THE MINE

Reports on the Mount Gunson mine have appeared in the South Australian Mining Reviews from time to time. From these reviews the following short resume has been compiled.

The copper deposits were discovered about the year 1895. In 1899, 50 tons of 14% ore are reported to have been dispatched for treatment. In 1901 the Government Geologist inspected the property and favourably reported on the copper occurrence. A reverberatory smelter was erected in 1905, but smelting operations were not successful on account of the highly siliceous nature of the ore. In 1909, Inspector Jones of the Mines Department recommended the erection of a treatment plant at the mine, as he then considered that development work was proving a highly promising deposit of copper. Between 1911 and 1915, the Mount Gunson Company worked the deposit and carried out testing operations prior to the erection of a suitable leaching plant. There is no record of the leaching plant having been completed or put into operation.

During 1917, 1918 and 1919, the company sent 1411 tons of dressed ore to smelters at Cockie Creek; 553 tons of the above ore were estimated to contain 8% copper. The Mount Gunson Company finally ceased operations in 1919.

RECENT OPERATIONS

In 1937, a mineral lease embracing the Mount Gunson and adjacent deposits was taken up by Mr. M.A. Mawby for the purpose of examining and testing the prospects thereon. The exploratory work carried out led up to the present mining operations.

Independent reports on the lease were made by Messrs. H.J.C. Conolly and J.K. Gustafsen for the Zinc Corporation Ltd. These reports have been made available to the Mines Department and data contained therein are incorporated in this report.

POSSIBLE ORE

The ore deposits at Mount Gunson are essentially low grade deposits and a large tonnage of possible ore must therefore be proved in them, before they can be considered as mining propositions. With a mill-head grade of 3.14%, Dr. J.K. Gustafsen made a possible tonnage estimate of 395,000 tons at 3.14%. This estimate includes figures for Pernatty Lagoon and Gunyot deposits. He has summarized the data on which the estimates was based in the following table:

TABLE 1.

ORE POSITION

as at 27th. July, 1937.

WORKINGS, ORE BLOCKS	Probable Ore		Probable and possible Ore		Remarks	
	Tons	Millhead Grade in %	Tons	Millhead Grade in %	Probable Ore	Probable & possible ore
1. Pernatty Lagoon	127,000	2.86	203,000	2.86	Wainwright 1914	Some Cu in quartzite below "leaching sand"
2. Gunyot	16,000	2.30	21,000	2.30	Wainwright 1914	Possible westward extension.
M (a) Main open cut	15,000	5.13	19,700	5.13	Practically proved	Limited by topography
t. (b) Tableland shaft	2,750	3.00	49,900	3.00	Fairly conservative	Assumes continuity half way to (d) & (e)
G (c) Western open cut & tunnel	510	6.75	510	6.75	Proved	Proved.
U (d) Young's open cut	3,430	6.10	25,800	4.00	Practically proved	Assumes continuity half way to (b)
N (e) Brennan's Workings	6,170	3.46	45,600	3.46	Assumes continuity between old & new pit	Assumes continuity half way to (b) and (c)
S (f) Main of Main Open Cut			20,600	3.00	Doubtful	Now being tested, grade assumed.
O (g) Smelter Shaft	1,780	3.00	1,780	3.00	Guesswork, old date limited; important position	
(h) Brennan's open cut	2,740	4.47	5,480	4.47	Practically proved	Assumes limited extension
N (j) Dumps	1,500	2.50	1,500	2.50	Roughly but conservatively estimated.	
All workings	177,350	3.12	394,870	3.14	Possibility of considerably extending these figures depends on: (1) Western extension of ore beyond (e): (2) Extension of smelter shaft and Brennan Shaft ore horizons.	
	177,000	say 3.12	395,000	say 3.14		
Open cut ore	171,000		330,000		Roughly estimated.	
Stope Ore	6,000		65,000		Most new ore not included in these estimates would probably be stoped.	

The estimate of 395,000 tons at 3.14% is insufficient to warrant the erection of a plant at the present price of copper (£86.10.0.). This estimate can only be increased by an exploratory campaign. The problem of the advisability of future exploration is dealt with at the end of the report.

Present ore position.

The present mining operations are concerned solely with the working of the higher grade patches of ore which can be profitably treated at the Port Pirie smelters. This ore is exposed in the main open cut.

Production, at present, amounts to about 200 tons or ore per week of copper (£86.10.0) is (£3.17.9) per ton.

Ore reserves stand at roughly 18,000 tons, assuming a continuation of the present system of open cut mining and a minimum grade of 4½%.

The maximum rate of production would be 250 tons per week, which is the maximum weekly tonnage that can be treated at Port Pirie.

The mine, on the above basis, has a life of 18 months.

Exploration and future prospects.

It is fairly definite that a profitable mine cannot be built up on the low-grade ore at present available in the open cuts. It is also very problematical whether it is advisable to undertake further exploration because of the following difficulties:-

- (1) Ore reserves of 18,000 tons at 4½% are not likely to yield a sufficient profit to pay for an exploratory campaign.
- (2) Any exploratory campaign can only be expected to yield low grade ore comparable to that already found.
- (3) Any exploratory campaign would involve the proving of a large tonnage of possible ore and also a considerable expense as extensive areas would require testing.
- (4) Open cut ore prospects are limited.
- (5) ~~Stope~~ Stope ore prospects are handicapped by the universal low-grade character of the copper seams and their relatively small widths.

(5)

(6) The majority of the actual prospects are concerned with stope ore for which mining costs would be high.

On the evidence available, the conclusion is that the deposits do not show sufficient promise to be worth further exploration work at present metal prices.

The exhaustion of the present ore reserves in the open cuts will bring the present mining operations at Mount Gunson to an end.

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